

3. Dictionaries and JSON

What you will be able to do:

- Explain how hashing works
- Perform basic operations on Dictionaries
- Parse JSON to generate interesting output

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Review (APIs)

poll



Active poll

0

Who supplies the API needed to access a resource?

- the Internet
- the client
- the server
- HTTP

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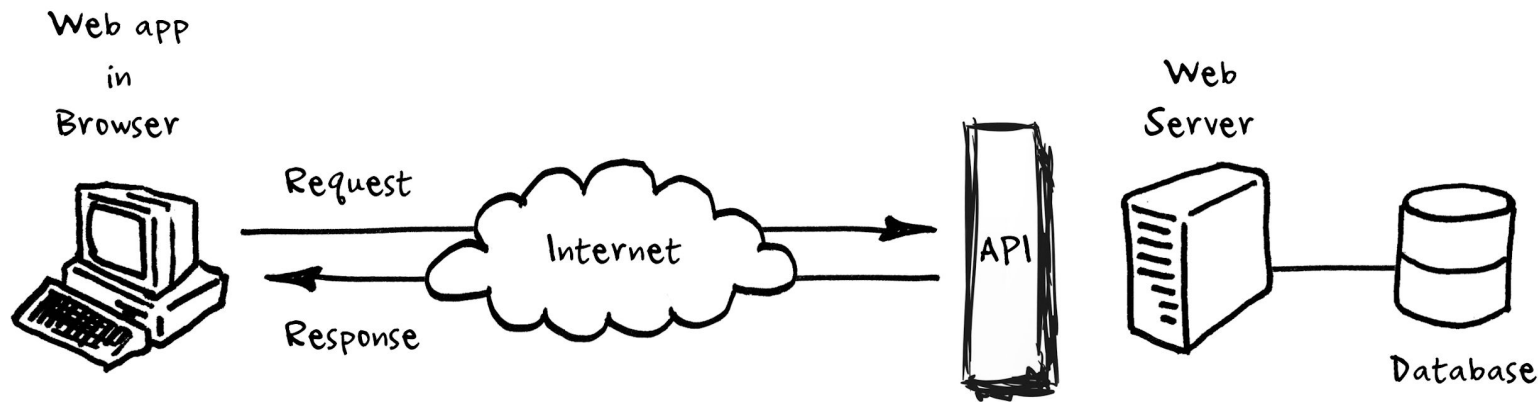
Definitions

- **Application Programming Interface (API)** - set of rules structuring interaction between applications
- **Library** - set of related, reusable code (e.g. pandas, matplotlib)
- **Framework** - structured code that makes it easier for a programmer or developer to create a desktop/mobile/web application; it usually includes a set of libraries to perform various tasks
- **REST** - most popular type of API; an architectural style
- **SOAP** - more secure version of REST
- **Software Development Kit (SDK)** - set of tools which can include libraries, APIs, frameworks, etc.

How APIs work

To access and interface with a site's stored information, we need to use their provided APIs:

1. **Client** sends a **request** for resources using an API **endpoint** , which includes a **URL** and **parameters**
2. **Server** sends **response** with the **resource**



DEMO Using APIs

Definitions

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Dictionaries

Do you know the following definitions?

- Hashtable
- Hashing function
- Dictionary
- Key-Value Pair

Definitions

Hash table – a data structure used to implement a dictionary as a means to allow quick look up keys to corresponding values

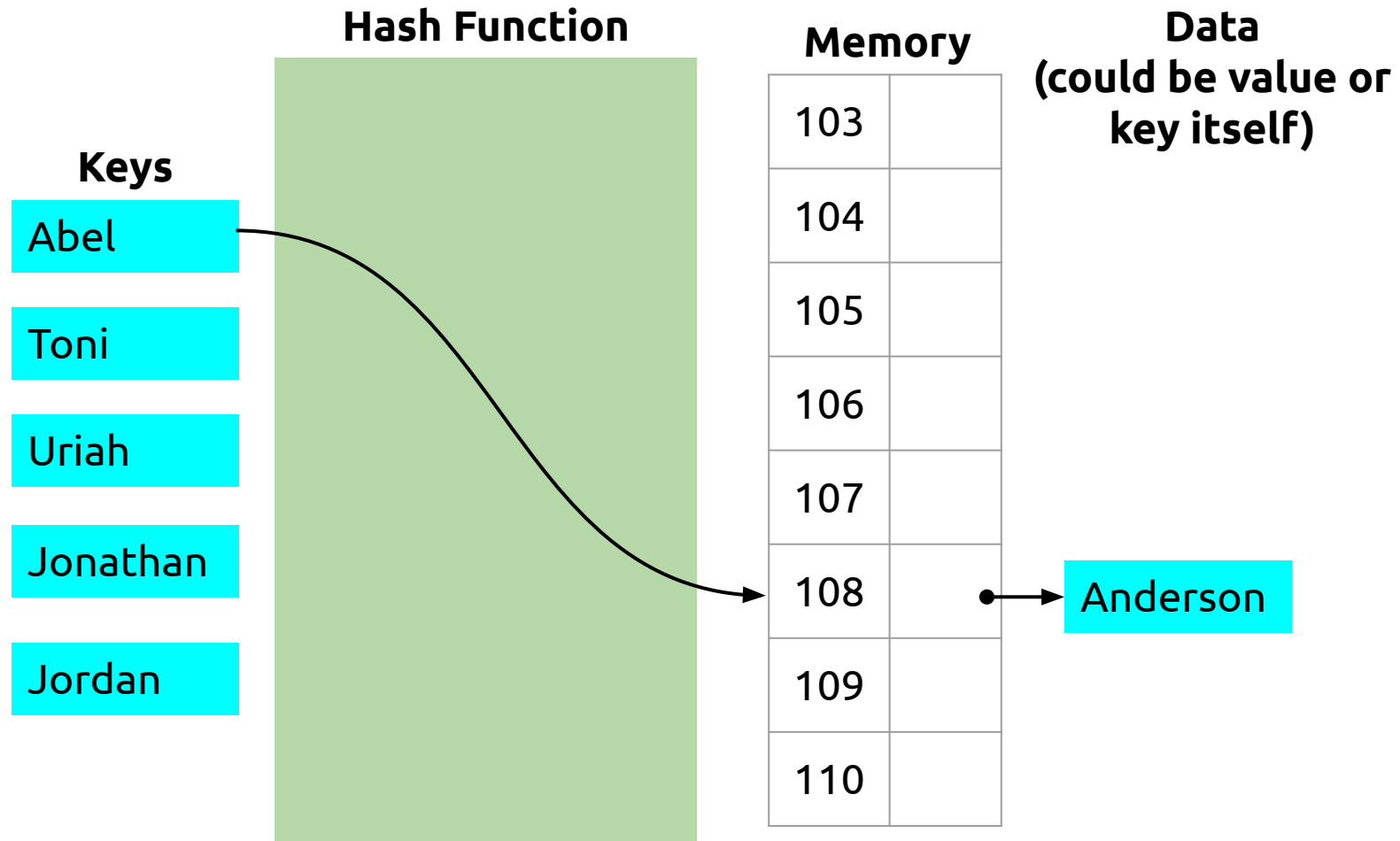
Hashing function – a one-way function often used to transform data into a random, formatted value. It can be used for encryption or indexing hashmaps or hash tables

Dictionary – the concept of storing data using a key and value system

Key-Value Pair – a hash table entry

- Fun fact, C# has Hashtables and Dictionaries as data structures - so take many of these definitions as high level

Hashing for Hashmaps and Tables....



Python: Using Dictionaries

- Say you want to map your zipcode (key) to a city using the syntax #####: 'city name',

- example:

77478: 'Sugar Land',

- A way to create a list of key-value pairs in Python is:

```
zip_codes = { 11201: 'Brooklyn',  
              94112: 'San Francisco' }
```

- Accessing information in dictionary:

```
print(zip_codes[11201])
```

Python: Modifying a Dictionary

```
dictionary['key'] = new_value
```

- creates/update a value:

```
pop(key)
```

- removes specified key and returns associated value

```
popitem()
```

- removes last item and returns tuple

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JSON

JavaScript Object Notation (JSON)

- Looks (and acts) like a **dictionary** with nested key-value pairs and lists
- Is sometimes returned as an API **GET** response
- Is easily parse-able by machines and systems
 - Python has **json** library with includes a function called **json.loads** that turns JSON into a dictionary data structure for easy manipulation!

```
{  
  "firstName": "John",  
  "lastName": "Smith",  
  "age": 21,  
  "address": {  
    "streetAddress": "21 2nd Street",  
    "city": "New York",  
    "state": "NY",  
    "zipCode": "10021"  
  },  
  "children": [  
    "Mary",  
    "James"  
  ]  
}
```


JSON-ifying in Python

```
import requests
import json

url = "https://api.genius.com/search?q=Kendrick%20Lamar"

my_headers = { "Authorization": "Bearer ACCESS_TOKEN" }

response = requests.get(url, headers=my_headers)
response_dict = json.loads(response.text);

print(response_dict["meta"]["status"]) // prints 200
```

```
codio@brazilindigo-gurusantana:~/workspace$
{"meta":{"status":200},"response":{"hits":[{"highl
ights":[],"index":"song","type":"song","result":{"
annotation_count":20,"api_path":"/songs/3039923","
artist_names":"Kendrick
Lamar","full_title":"HUMBLE. by Kendrick
Lamar","header_image_thumbnail_url":"https://image
s.genius.com/483306c535608c27f9e3781b854dc91d.300x
300x1.png","header_image_url":"https://images.geni
us.com/483306c535608c27f9e3781b854dc91d.1000x1000x
1.png","id":3039923,"lyrics_owner_id":104344,"lyri
cs_state":"complete","path":"/Kendrick-lamar-humbl
e-lyrics","pyongs_count":1203,"relationships_index
_url":"https://genius.com/Kendrick-lamar-humble-sa
mple","release_date_components":{"year":2017,"mont
h":3,"day":30},"release_date_for_display":"March
30....
```

JSON Syntax, the basics

- Maps are denoted with `{}` (technically, these are called objects)
 - `{"a": 1, "b": 2, "c": 3}` is a map for a to 1, b to 2, and c to 3
 - JSON *always* starts with a '{' and ends with an '}'
- A few basic data types: numbers, strings, booleans, and nulls
- These types work as you would expect in any programming language
- Arrays using `[]`
- `[1, 2, 3, 4, 5]` is an array of 1, 2, 3, 4, and 5

```
{
  "firstName": "John",
  "lastName": "Smith",
  "age": 21,
  "address": {
    "streetAddress": "21 2nd Street",
    "city": "New York",
    "state": "NY",
    "zipCode": "10021"
  },
  "children": [
    "Mary",
    "James"
  ]
}
```

JSON Syntax, nesting

- Nested arrays and maps can live recursively inside each other, and vice versa
- `[{"name": "a"}, {"name": "b"}, {"name": "c"}]`
is an array of maps, each one with a key called name
- `{"users": [{"name": "a"}, {"name": "b"}, {"name": "c"}]}`
is a map with one key called users, whose value is an array of maps (described above)

JSON Parsing

```
{  
  "isStale": false,  
  ...  
  "number": 10,  
  "results": [  
    {  
      "title": "Kale and Quinoa Salad with Black Beans",  
      ...  
    },  
    {  
      "title": "Citrus Sesame Kale",  
      ...  
    }  
  ]  
}
```

```
json_body["results"]
```

JSON Parsing

```
{  
  "isStale": false,  
  ...  
  "number": 10,  
  "results": [  
    {  
      "title": "Kale and Quinoa Salad with Black Beans",  
      ...  
    },  
    {  
      "title": "Citrus Sesame Kale",  
      ...  
    }  
  ]  
}
```

```
json_body["results"][0]
```

JSON Parsing

```
{  
  "isStale": false,  
  ...  
  "number": 10,  
  "results": [  
    {  
      "title": "Kale and Quinoa Salad with Black Beans",  
      ...  
    },  
    {  
      "title": "Citrus Sesame Kale",  
      ...  
    }  
  ]  
}
```

```
json_body["results"][0]["title"]
```

DEMO Parsing JSON

Reminders

- Check your Google Calendars for this week's events!

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Thank you!